

Ankit Joshi, Ph.D.

Phone: +1 (224) 266-3719 | Email: ajoshi@bhsai.org | [linkedin.com/in/joshivank/](https://www.linkedin.com/in/joshivank/) | [Google Scholar Profile](https://scholar.google.com/citations?user=ajoshi)

SUMMARY

My research focuses on human-centered thermal science that integrates human-environment interactions, thermal physiology, and engineering to quantify and mitigate adverse impacts of heat on health, comfort, and safety. My work spans experimental studies, the development of theoretical and computational frameworks to explain observed thermal responses, and the translation of this knowledge into field-ready applications. For example, I have advanced urban microclimate measurement systems to quantify heat exposure, developed human thermoregulation models to uncover physiological mechanisms underlying heat-related health risks, and evaluated mitigation strategies such as reflective surfaces, shading, and personal cooling technologies. Overall, I have a proven record of translating mechanistic research into deployable design tools and performance-driven insights that strengthen thermal resilience, public-health protection, and human well-being.

EXPERIENCE

Research Scientist II

May. 2025 – Present

Henry M. Jackson Foundation for the Advancement of Military Medicine (HJF), Frederick, MD, United States
(In support of the Department of Defense Biotechnology High Performance Computing Software Applications Institute)

- Leading development of the **first high-fidelity female thermoregulation model** to improve prediction of heat strain and injury risk in military personnel, integrating anatomical, hormonal, and physiological variability.
- Designing and conducting validation studies using controlled chamber and field datasets to **achieve < 0.2 °C mean error** in predicted core temperature across diverse activity levels and environments.
- Developing COMSOL- and MATLAB-based frameworks to **quantify sex-specific thermal responses** and translate findings into **decision-support tools** for DoD heat and cold-stress management.

Research Scientist

Feb. 2023 – May. 2025

Arizona State University, Tempe, AZ, United States

- Developed **advanced methods to quantify human extreme heat exposure** using thermal manikins and biometeorological instruments, advancing the understanding of human-environment heat exchange
- Developed **thermoregulation models with varying complexity** from simplified lumped parameter (MATLAB) to advanced 3D anatomical model (FEA), achieving high accuracy in predicted core temperature (Error $\pm 0.3^{\circ}\text{C}$)
- Computationally analyzed the **impact of human body shape** on heat transfer using **COMSOL Multiphysics**
- Conducted **15+** human subject trials to analyze the **microscale human sweating mechanism** and evaporation
- Engineered industry-grade sensors for measuring thermal radiation and wind speed, reducing costs by **95%**
- Published **10+** peer reviewed research articles and presented research outcomes in **4+** international conferences
- **Mentored 15+** undergraduate, graduate, and PhD students; **taught** courses on heat transfer topics
- Collaborated with a **multidisciplinary team** of urban biometeorologists, physiologists, & engineers

Postdoctoral Fellow

Sep. 2021 – Feb. 2023

Iowa State University, Ames, IA, United States

- Developed **3D hand thermoregulation model for cold weather injuries (ANSYS)**, optimizing protective design
- Developed & validated a **computational tool to design PPEs (e.g., firefighter gloves)**, optimizing protection, dexterity, & safe working durations for thermal comfort and cold-injury prevention in extreme environments
- Co-authored a **successful NIOSH grant [\$731,250]** and two NSF/NIH grant proposals to address cold strain
- **Authored** a peer reviewed research article, 3 book chapters, and mentored 2 graduate students

Research Collaboration (Remote)

May. 2020 – Apr. 2021

- Developed an advanced 3D thermoregulation model integrating realistic and complex clothing, using **ANSYS** and **C** to enhance accuracy and reliability of thermoregulation models in diverse thermal environments
- Published a peer-reviewed article demonstrating robust and precise human thermoregulation modeling outcomes, with **high accuracy** in predicted core and skin temperature of $\pm 0.3^{\circ}\text{C}$ and $\pm 0.6^{\circ}\text{C}$ respectively

PhD candidate

Jan. 2017 – Jan. 2020

Swiss Federal Laboratories for Materials Science and Technology (Empa), Switzerland

- Developed comprehensive model for heat & mass transfer through clothing to enhance thermal comfort & protection, achieving accuracy and reliability (Mean error <10%) of experimental methods like thermal manikin
- Published 3 peer-reviewed articles, presented at 3+ international conferences, and forged industry collaborations

Intern and Master thesis

Apr. 2015 – Oct. 2015

SMS group GmbH, Düsseldorf, Germany

- Led large eddy simulations (**LES**) of supersonic oxygen injectors and validated the results against **DNS**
- Improved injector design to achieve a **42%** longer supersonic jet to enhance oxygen efficiency

EDUCATION

Ph.D. in Mechanical Engineering

Jan. 2017 – Dec. 2019

Ecole Nationale Supérieure d'Ingénieurs Sud Alsace, France

Thesis title: "Comprehensive model for heat and moisture transport in skin-clothing-environment system."

Master of Science in Computational Mechanics

Oct. 2013 – Oct. 2016

University of Duisburg-Essen, Germany

Thesis title: "Large Eddy Simulation of supersonic oxygen injector for the Electric Arc Furnace"

ACADEMIC SERVICES

Peer Reviewer for Scholarly Journals & Funding Agencies

Dec. 2021 - Present

Reviewed 30+ articles & grant proposals for:

- National Science Foundation
- Scientific Reports
- Journal of Thermal Biology
- Building and Environment
- Applied Ergonomics
- International Journal of Occupational Safety and Ergonomics
- Royal Society Open Science

Co-chair, 12th International Manikin and Modelling Meeting (12i3m), Switzerland

Aug. 2018

GRANTS & FUNDING

Centers for Disease Control and Prevention

Sep. 2023

National Institute for Occupational Safety and Health (NIOSH), Project #1U01OH012573-01,

A systematic approach to enhance extremity protection in the Alaska fishing industry. [Award Amount: \$731,250]

Postdoctoral Seed Grant

Jun 2022

Iowa State University, USA [Award Amount: \$2,000]

KEY SKILLS

FEA Tools:	COMSOL Multiphysics, ANSYS Workbench, STAR-CCM+, OpenFOAM
Programming:	MATLAB, Python, and C
Collaboration & Mentorship:	Mentored 20+ undergraduate, graduate, and PhD students
Teaching:	Fundamental & Advanced Heat Transfer and Computational Fluid Dynamics

PEER REVIEWED PUBLICATIONS ([Google Scholar](#))

1. **Joshi, A.**, Twidwell, B., Park, M., & Rykaczewski, K. (2025). Comparative analysis of thermoregulation models to assess heat strain in moderate to extreme heat. *Journal of Thermal Biology*, 127, 104035. <https://doi.org/10.1016/j.jtherbio.2024.104035>
2. Jose, C. T., **Joshi, A.**, Viswanathan, S. H., Nash, S. K., Sadeghi, K., Kavouras, S. A., & Rykaczewski, K. (2025). A micro-to-macroscale and multi-method investigation of human sweating dynamics. *Journal of the Royal Society Interface*, 22(228), 20250407. <https://doi.org/10.1098/rsif.2025.0407>
3. Sadeghi, K., Viswanathan, S. H., **Joshi, A.**, Bartels, L., Wereski, S., Jose, C. T., ... & Rykaczewski, K. (2025). Resolving shortwave and longwave irradiation distributions across the human body in outdoor built environments. *Building and Environment*, 277, 112934. <https://doi.org/10.1016/j.buildenv.2025.112934>

4. Rykaczewski, K., **Joshi, A.**, Viswanathan, S. H., Parkerson, E., Gupta, M., Park, M., ... & Middel, A. (2025). Advanced human heat exposure sensing using two cylinder anemometer and radiometer: introducing CARla. *International Journal of Biometeorology*, 1-14. <https://doi.org/10.1007/s00484-025-02860-4>
5. Viswanathan, S. H., **Joshi, A.**, Bartels, L., Sadeghi, K., Vanos, J. K., & Rykaczewski, K. (2025). Impact of human body shape on free convection heat transfer. *PloS one*, 20(2), e0318842. <https://doi.org/10.1371/journal.pone.0318842>
6. **Joshi, A.**, Viswanathan, S. H., Jaiswal, A. K., Sadeghi, K., Bartels, L., Jain, R. M., ... & Rykaczewski, K. (2024). Characterization of human extreme heat exposure using an outdoor thermal manikin. *Science of the Total Environment*, 923, 171525. <https://doi.org/10.1016/j.scitotenv.2024.171525>
7. Vanos, J. K., **Joshi, A.**, Guzman-Echavarria, G., Rykaczewski, K., & Hosokawa, Y. (2024). Impact of Reflective Roadways on Simulated Heat Strain at the Tokyo, Paris and Los Angeles Olympics. *Journal of Science in Sport and Exercise*, 6(3), 288-302. <https://doi.org/10.1007/s42978-024-00294-9>
8. Karanja, J., Vanos, J., **Joshi, A.**, Penner, S., Guzman, G. E., Connor, D. S., & Rykaczewski, K. (2024). Impact of tent shade on heat exposures and simulated heat strain for people experiencing homelessness. *International journal of biometeorology*, 1-14. <https://doi.org/10.1007/s00484-024-02751-0>
9. Jaiswal, A. K., Jose, C. T., Ramesh, R., Nanani, V. K., Sadeghi, K., **Joshi, A.**, ... & Rykaczewski, K. (2024). Simultaneous imaging of multi-pore sweat dynamics and evaporation rate measurement using wind tunnel ventilated capsule with infrared window. *iScience*. <https://doi.org/10.1016/j.isci.2024.110304>
10. Rykaczewski, K., **Joshi, A.**, Viswanathan, S. H., Guddanti, S. S., Sadeghi, K., Gupta, M., ... & Middel, A. (2024). A simple three-cylinder radiometer and low-speed anemometer to characterize human extreme heat exposure. *International journal of biometeorology*, 1-12. <https://doi.org/10.1007/s00484-024-02646-0>
11. **Joshi, A.**, Li, R., Wu, Y., Zhang, M., & Song, G. (2024). Analysis of glove local microclimate properties for various glove types and fits using 3D scanning method. *Heliyon*, 10(1). <https://doi.org/10.1016/j.heliyon.2023.e23596>
12. **Joshi, A.**, Bartels, L., Viswanathan, S. H., Martinez, D. M., Sadeghi, K., Jaiswal, A. K., ... & Rykaczewski, K. (2023). Evaluation of thermal properties and thermoregulatory impacts of lower back exosuit using thermal manikin. *International Journal of Industrial Ergonomics*, 98, 103517. <https://doi.org/10.1016/j.ergon.2023.103517>
13. **Joshi, A.**, Psikuta, A., Annaheim, S., & Rossi, R. M. (2023). Modelling of heat and mass transfer in clothing considering evaporation, condensation, and wet conduction with case study. *Building and Environment*, 228, 109786. <https://doi.org/10.1016/j.buildenv.2022.109786>
14. **Joshi, A.**, Wang, F., Kang, Z., Yang, B., & Zhao, D. (2022). A three-dimensional thermoregulatory model for predicting human thermophysiological responses in various thermal environments. *Building and Environment*, 207, 108506. <https://doi.org/10.1016/j.buildenv.2021.108506>
15. **Joshi, A.**, Psikuta, A., Bueno, M. A., Annaheim, S., & Rossi, R. M. (2021). Effect of movement on convection and ventilation in a skin-clothing-environment system. *International Journal of Thermal Sciences*, 166, 106965. <https://doi.org/10.1016/j.ijthermalsci.2021.106965>
16. **Joshi, A.**, Psikuta, A., Bueno, M. A., Annaheim, S., & Rossi, R. M. (2019). Analytical clothing model for sensible heat transfer considering spatial heterogeneity. *International Journal of Thermal Sciences*, 145, 105949. <https://doi.org/10.1016/j.ijthermalsci.2019.05.005>
17. **Joshi, A.** (2019). *Comprehensive model for heat and mass transfer in human skin-clothing-environment system* (Doctoral dissertation, Université de Haute Alsace-Mulhouse). [Link to Thesis](#)

BOOK CHAPTERS

18. **Joshi, A.**, & Zhang, M. (2025). Respiratory protective device: Protection mechanisms, filtration efficiency, and associated modeling and predictions. In *Designing Advanced Respiratory Protective Devices for Pandemics* (pp. 115-135). Elsevier Science Ltd. <https://doi.org/10.1016/B978-0-323-95316-0.00011-6>
19. Li, R., **Joshi, A.**, Wu, Y., & Zhang, M. (2025). Purpose and performance of respiratory protective devices in a pandemic. In *Designing Advanced Respiratory Protective Devices for Pandemics* (pp. 17-56). Elsevier Science Ltd. <https://doi.org/10.1016/B978-0-323-95316-0.00003-7>
20. Gao, H., Li, R., & **Joshi, A.** (2025). Future perspective and outlook: Incorporation of advanced technology. In *Designing Advanced Respiratory Protective Devices for Pandemics* (pp. 289-313). Elsevier Science Ltd. <https://doi.org/10.1016/B978-0-323-95316-0.00001-3>

PUBLICATIONS IN REVIEW

21. Abdullah, M., Viswanathan, S., Jose, C. T., ..., Rykaczewski, K & **Joshi, A.** (2025). Turbulence of airflow matters in human thermophysiological response.

CONFERENCES

1. **Joshi, A.**, Viswanathan, S. H., Jose, C. T., Pathikonda, G., & Rykaczewski, K. (2025, October). *Quantifying turbulent convective heat transfer using an outdoor thermal manikin*. In *Proceedings of the 13th International Manikin & Modelling Meeting (I3M)*
2. **Joshi. A.**, Rykaczewski, K., "Thermoregulation Models for Assessing Heat Strain in Extreme Conditions," 16th Conference on Environment and Health, January 12-16, 2025, New Orleans, Louisiana, USA.
3. **Joshi, A.**, Viswanathan, S., Jaiswal, A., Sadeghi, K., Bartels, L., Pathikonda, G., & Rykaczewski, K. (2024, November). Forced convection from the human body in turbulent outdoor flows. In APS Division of Fluid Dynamics Meeting Abstracts (pp. J32-003).
4. **Joshi. A.**, Wang. F., "Predicting human thermal response using 3D thermoregulation model," International Congress of Biometeorology, May 14-17, 2023, Tempe, Arizona, USA.
5. **Joshi. A.**, Psikuta. A., Bueno. M.-A. , Annaheim. S. and Rossi. R., "Realistic clothing model for heat and moisture transport through human skin-clothing-environment system," International Conference on Environmental Ergonomics, 7-12 July 2019, Amsterdam, The Netherlands.
6. **Joshi. A.**, Psikuta. A., Bueno. M.-A. , Annaheim. S. und Rossi. R., "Analysis Of Heat Transfer From Human Body And Effect Of Clothing Surface On Heat Transfer Mechanism", COMSOL conference, 22-24 Oct 2018, Lausanne, Switzerland.
7. **Joshi. A.**, Psikuta. A., Bueno. M.-A. , Annaheim. S. und Rossi. R., Mathematical formulation of sensible heat transfer in the spatially heterogeneous skin-clothing-environment system, 12th International Manikin and Modelling Meeting (12i3m), 29-31 Aug 2018, St. Gallen, Switzerland

WEBINAR PRESENTATION

8. M. Zhang, **Joshi. A.**, R. Li, Y. Wu, and G. Song, A novel approach for determining glove area factors based on the 3D scanning technology, 12th 3DBODY.TECH Conference & Expo, Oct 2021.